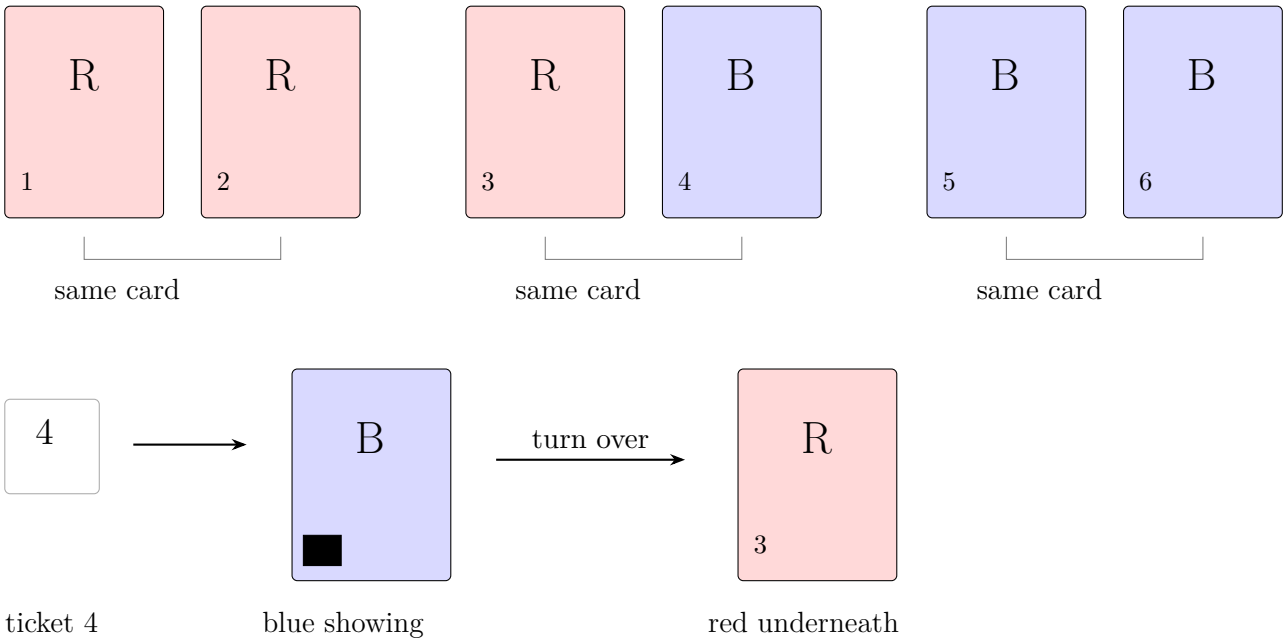
The small numbers match the six tickets. A secret ticket picks the face that shows, with its number covered. You see only the color. Every ticket has the same chance.

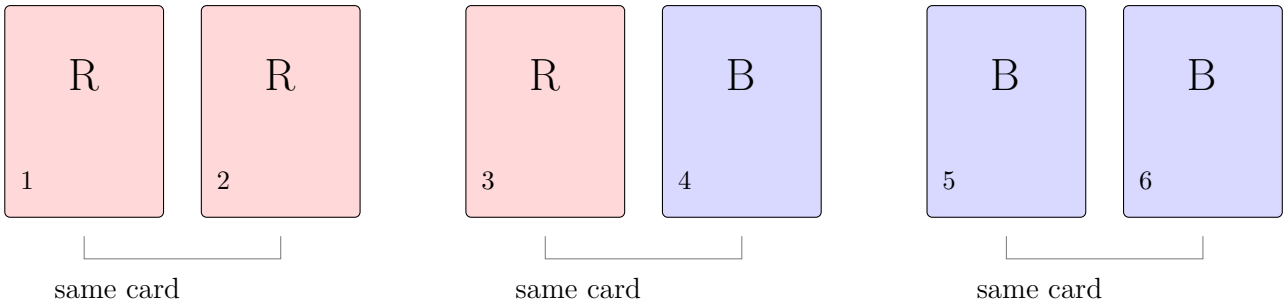


Problem 1: Given only a red showing color, determine the chances of the hidden colors exactly. Do the same for a blue showing color.

red showing

blue showing

Problem 2: A friend says, “Red is showing, so only the red/red and red/blue cards remain. Each is one card, so each must have the same chance.” Does the selection rule support that argument?



Problem 3: The chooser also tells you one fact about the covered number: the number, whether it is odd or even, or whether it is 3 or less. After a red clue, what does each fact do to the better guess?

the number

odd or even

3 or less

Only tickets 1 and 3 are in the cup, each with the same chance.



Problem 4: Does this way of producing a red clue change the chance that red is hidden? Give an exact comparison with the six-face-ticket rule.

Problem 5: Could six red-showing rounds prove that the hidden-color chances differ?

Use a cup containing a subset of the original six tickets, with no repeats. Each selected ticket has the same chance.

Problem 6: Find every subset that makes red showing equally likely to hide red or blue. Your cup must sometimes show red; it may also show blue.

1	2	3	4	5	6

Problem 7: Can you keep only whole cards and make that red-clue game fair, with red still possible? Keep one, two, or all three cards, and remove both tickets whenever a card goes.